

MUHAMMAD IBTISAM TANVEER

Germany

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Profile Summary

Frontend Software Engineer with experience building scalable web applications using **Next.js**, **React**, **TypeScript** and **Tailwind CSS**. Skilled in implementing complex user flows, integrating backend APIs, and designing maintainable frontend architectures. Passionate about creating robust, performant, and user-focused web applications.

Education

Technical University Of Chemnitz

Masters in Web Engineering

2024 – 2026

Chemnitz, Saxony

National University Of Technology

Bachelor of Science in Computer Science

2018 – 2022

Islamabad, Pakistan

Experience

Smart Energy Pays

Software Engineer (Frontend-Focused)

Feb 2025 – Present

Berlin, Germany

- Developed a scalable banking web platform supporting retail, business, and junior products using **Next.js 15 App Router**, **React 19**, **TypeScript**, and **Tailwind CSS**.
- Designed and structured a maintainable frontend architecture with route groups, shared layouts, and dynamic routes for dashboards and product modules.
- Implemented complex user flows including **KYC/KYB onboarding**, **crypto wallet management**, **transfers**, **subscriptions**, and **support ticketing** based on product designs.
- Integrated backend services through a centralized **API fetching layer with retry logic and token refresh**, ensuring reliable communication with authentication, payment, and business APIs.
- Managed application state and server data using **Redux Toolkit** and **TanStack React Query** for predictable state management and efficient data fetching.
- Collaborated with backend and product teams to define data contracts, implement validation and error handling, and support real-time updates via **WebSockets**.
- Optimized application performance and developer experience using **React Query caching**, **Turbopack**, and **next-intl** for internationalization.

Gigalabs (Pvt) Ltd.

Software Engineer

Nov 2022 – Sep 2024

Lahore, Pakistan

- Developed full-stack features across multiple production products including **PIMdesk**, **Pushka Hub**, and enterprise integrations using **React**, **Next.js**, **TypeScript**, **Node.js**, and **NestJS**.
- Built scalable UI components, dashboards, and data-management interfaces for **PIMdesk**, a Product Information Management platform used to centralize and manage large e-commerce product datasets.
- Implemented backend services and **RESTful APIs using NestJS and Flask**, enabling reliable data synchronization between frontend applications, databases, and third-party systems.
- Developed data processing workflows for importing, managing, and exporting large product catalogs, improving efficiency in managing multi-channel e-commerce data.
- Refactored legacy JavaScript codebases into **TypeScript**, improving maintainability, type safety, and reducing runtime errors in production systems.
- Designed and optimized **PostgreSQL** database queries and schemas to support large product and transactional datasets.
- Collaborated with product managers, designers, and backend engineers to debug production issues, implement new features, and improve system reliability.
- Enhanced frontend performance and responsiveness through component refactoring, optimized state management, and efficient API integration.

Projects

WAM Studio (Enfield) – Web Architecture Diagram Editor | *Next.js, React 19, TypeScript, Tailwind CSS, React Flow*

- Developed a collaborative web-based diagram editor for Web Architecture Modeling (WAM) using **Next.js App Router** and **React 19**.
- Implemented core UI components including diagram canvas, node palette, properties panel, version history, and export tools using **React Flow**.
- Integrated REST APIs for diagram CRUD operations, sharing, comments, and version management with structured data flow and error handling.
- Designed a maintainable feature-based frontend architecture with typed APIs, reusable hooks, and React context for authentication and global state.
- Enabled real-time collaboration and export functionality (JSON, PNG, PDF) to support architecture documentation and team workflows.

Pushka Hub | *React Native, Ruby on Rails*

- Developed cross-platform mobile features using **React Native**, improving application responsiveness and performance.
- Implemented secure wallet and transaction flows while integrating backend APIs built with **Ruby on Rails**.
- Enhanced UI/UX across Android and iOS platforms, improving usability and user engagement.
- Debugged and optimized application stability, reducing crash rates and improving reliability.

PIMdesk | *Vue.js, NestJS, Ruby on Rails, Shopify*

- Developed frontend interfaces using **Vue.js** for a Product Information Management platform managing large e-commerce product catalogs.
- Integrated APIs enabling bulk product uploads, dynamic attribute mapping, and product image management.
- Implemented advanced search, filtering, and tagging features to streamline product discovery and organization.
- Optimized platform performance for efficient management of large product datasets across multiple sales channels.

AAT Wholesale E-Commerce App | *React Native, Laravel*

- Developed a cross-platform wholesale e-commerce mobile application using **React Native**.
- Implemented product catalog management supporting **10,000+ SKUs** with search, filtering, and category-based navigation.
- Enhanced user experience through wishlists, push notifications, and review systems to improve engagement.

Consumer Information App (SNGPL) | *Flutter, Provider*

- Developed a mobile application for meter reading and consumer complaint management used by field teams.
- Integrated location services to support field operations and improve response time for service requests.
- Achieved **50,000+ downloads** on the Google Play Store.

Automated Personality Prediction System | *Python, TensorFlow, Flask, OpenCV*

- Built a machine learning system to predict OCEAN personality traits using deep learning models.
- Processed and analyzed **10,000+ videos** using computer vision techniques.
- Integrated the ML model with a Flask API to provide real-time predictions for the mobile application.

Technical Skills

Programming Languages: JavaScript (ES6+), TypeScript, Python, SQL

Frontend: React, Next.js (App Router), Vue.js, Redux Toolkit, React Query, Tailwind CSS, Material UI, HTML5, CSS3/SCSS, Responsive Design, UI/UX Implementation

Backend: Node.js, Express.js, NestJS, Flask, Django, REST APIs, JWT Authentication

Databases: PostgreSQL, MongoDB

Developer Tools: Git, GitHub, GitLab, VS Code, Jira, Postman, Android Studio, Jupyter Notebook, Figma, Vercel, AWS (EC2, S3)

Machine Learning: TensorFlow, Scikit-learn, OpenCV, Pandas, Matplotlib

Practices: Clean Architecture, Design Patterns, Debugging, Code Review, Unit Testing, Agile/Scrum, Project Management (ClickUp)